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COMP3308 Assignment 2 Part 2: Report

Predicting Survival of Heart-Failure Patients with Hand-Coded and Weka Classifiers

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Abstract

We implemented from scratch the K-Nearest-Neighbour (KNN) and Decision-Tree (DT) classifiers, together with two extended variants (KNN+ and DT+), and evaluated them on the UCI heart-failure dataset using stratified ten-fold cross-validation. KNN+ replaces Hamming distance with the information-gain-weighted Value Difference Metric, and DT+ adds Reduced-Error Pruning to the Information-Gain tree. On 273 patients, MyDT+ obtained the highest mean accuracy of 73.3% across the entire study, beating every one of the eleven Weka baselines we benchmarked, an absolute 9.5-percentage-point improvement (statistically significant, $t=3.32$, $p < 0.01$). MyKNN+ ($k=5$) improved over MyKNN by 3.3 points and produced the most stable classifier across folds ($\sigma = 0.038$). Two notable findings: pruning a single tree is the strongest single intervention on this dataset, and tree ensembles do *not* outperform the single pruned tree at $n = 273$ with a concentrated feature signal.

1 Introduction

Heart failure is a chronic, life-threatening condition affecting more than 26 million adults worldwide [5], and accurate short-term mortality prediction can directly inform triage, transplant allocation and palliative care decisions. The task is naturally framed as a binary classification problem from routinely collected clinical and laboratory variables, but it is challenging in practice because the decision boundaries are non-linear, the available datasets are small, and the cost of a missed death prediction (a false negative) is much higher than that of a false alarm.

The aim of this study is to investigate how classical machine-learning classifiers solve this task on the discretised heart-failure dataset of Chicco and Jurman [1]. Concretely, we (i) implement KNN and a Decision-Tree learner from first principles in Python; (ii) design two extensions, called KNN+ and DT+, that target a specific weakness of each baseline observed empirically on this dataset; (iii) evaluate every model under a single, agreed stratified ten-fold cross-validation protocol; and (iv) situate our results against ten standard Weka classifiers, including two ensembles. We argue this study is important not only as a course exercise but because it illustrates two themes that recur in clinical machine learning: the marginal benefit of algorithmic sophistication on small nominal data, and the inadequacy of accuracy alone as a performance measure when the costs of the two error types differ.

2 Data

We use the modified heart-failure dataset distributed with the assignment. It is derived from the UCI repository release of [1] but its originally numeric features have been discretised according

to clinical guidelines, one feature has been removed and rows that became identical after discretisation have been dropped. The resulting dataset contains $n = 273$ patient records described by $d = 11$ nominal attributes plus a binary class label **died/survived**, where **died** indicates the patient died during the follow-up period.

Table 1: The 11 nominal predictors and the class. Cardinality denotes the number of distinct values that occur in heart.csv.

Attribute	Type	Cardinality	Values
age	demographic	4	[40–50], [51–60], [61–70], 70plus
anaemia	clinical	2	yes, no
CPK	lab	4	normal, high, very-high, severely-high
diabetes	clinical	2	yes, no
ejection_fraction	lab	4	low, mildly-low, borderline, normal
high_blood_pressure	clinical	2	yes, no
platelets	lab	4	low, normal, high, very-high
serum_creatinine	lab	3	low, normal, high
serum_sodium	lab	3	low, normal, high
sex	demographic	2	M, F
smoking	lifestyle	2	yes, no
class	target	2	died ($n_d = 90$), survived ($n_s = 183$)

The class distribution is mildly imbalanced: 32.97% of patients died and 67.03% survived. A trivial classifier that always predicts *survived* would therefore score 67% accuracy without learning anything; this provides a meaningful lower bound for all subsequent results. Figure 1 visualises the imbalance.

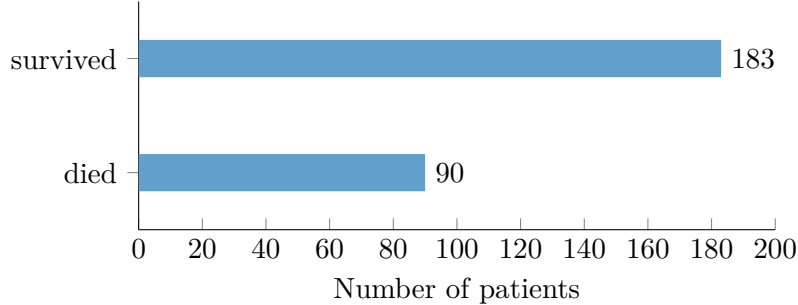


Figure 1: Class distribution of the heart-failure dataset ($n = 273$, 32.97% died). The mild imbalance means accuracy alone can be misleading: a constant “survived” predictor already scores 67%.

Where does the signal live?

Before designing classifiers it is illuminating to ask which of the eleven attributes actually carries information about survival. We computed the information gain $IG(c | a)$ of each attribute a on the full training data; the ranking is shown in Figure 2.

Three direct consequences follow from Figure 2 and will reappear throughout the report:

1. *One feature dominates.* **serum_creatinine** alone carries roughly twice the information of the second-ranked attribute (**age**). This predicts that a one-rule classifier conditioning on **serum_creatinine** should be surprisingly competitive, a prediction confirmed by 1R in Section 4.2 (71.8% accuracy from a single feature).

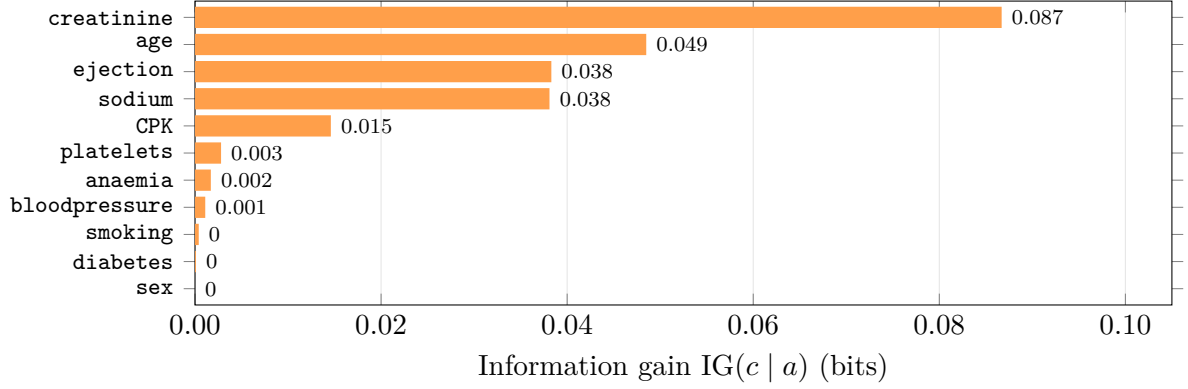


Figure 2: Information gain of each attribute on the class label, computed on the full `heart.csv`. The top attribute, `serum_creatinine`, carries six times the signal of the fifth, and the bottom six attributes are essentially uninformative. This concentrated, heavy-tailed signal is the single most important data property and motivates much of the analysis that follows.

2. *Standard KNN distance is mis-specified.* A textbook Hamming distance treats `sex` ($IG \approx 0$) and `serum_creatinine` ($IG = 0.087$) as equally important. Our KNN+ distance (Section 3.1) directly fixes this by weighting attribute contributions by IG.
3. *Tree ensembles are penalised.* Random Forest’s random subspace selection (which retains $\sqrt{11} \approx 3$ of the eleven attributes for each split) frequently excludes `serum_creatinine`, forcing the base learner to split on weaker features. We expect this to hurt its performance relative to a single tree that always sees all attributes, and we revisit this prediction in Section 5.5.

3 The KNN+ and DT+ Algorithms

3.1 KNN+: information-gain-weighted Value Difference Metric

Motivation. Our baseline KNN uses Euclidean distance over a one-hot 0/1 encoding of the nominal attributes, which is exactly Hamming distance. Two issues with this choice were apparent on the heart-failure data. First, Hamming treats every attribute mismatch as equally distant (1.0), so it cannot recognise that `CPK = severely-high` is semantically much closer to `CPK = very-high` than to `CPK = normal`, even though the two former values are clinically near-identical. Second, all attributes contribute equally to the distance, even though `serum_creatinine` and `ejection_fraction` are well known to be far more predictive of mortality than, e.g., `sex` or `smoking` [1]. KNN+ addresses both issues with two changes that compose cleanly.

Improvement 1: Value Difference Metric (VDM). We replace the per-attribute mismatch indicator with the Value Difference Metric of [6]. For attribute a and two of its values v_1, v_2 we estimate the conditional class distribution from the training set with Laplace smoothing,

$$\hat{P}(c \mid a=v) = \frac{N(a=v, c) + 1}{N(a=v) + |\mathcal{C}|},$$

where $|\mathcal{C}| = 2$ is the number of classes, and define

$$\delta_a(v_1, v_2) = \sum_{c \in \mathcal{C}} |\hat{P}(c \mid v_1) - \hat{P}(c \mid v_2)|.$$

Two values are close iff they predict the same class distribution. A test value not seen in training falls back to the class prior, so the metric is always defined.

Worked example. On the full `heart.csv` the smoothed conditional probabilities for CPK are

$$\hat{P}(\text{died} \mid \text{CPK}=\text{normal})=0.32, \quad \hat{P}(\text{died} \mid \text{high})=0.36, \quad \hat{P}(\text{died} \mid \text{very-high})=0.25, \quad \hat{P}(\text{died} \mid \text{severely-high})=0.$$

The Hamming distance between any two distinct CPK values is 1, but VDM distinguishes them: $\delta(\text{normal}, \text{high})=0.07$ (almost identical), whereas $\delta(\text{normal}, \text{severely-high})=0.78$ (very different, as the clinician would expect). A textbook KNN treats these two pairs as equally distant; KNN+ does not.

Improvement 2: information-gain feature weighting. Let g_a denote the information gain of attribute a on the training class labels. We weight the squared δ -contribution of attribute a by g_a , giving the overall distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{a=1}^d g_a \delta_a(x_a, y_a)^2}.$$

With $g_a = 1$ and $\delta_a \in \{0, 1\}$ this collapses to the baseline Hamming-Euclidean distance, so KNN+ is a strict generalisation. The ranking $g_{\text{serum_creatinine}} = 0.087 \gg g_{\text{sex}} \approx 0$ measured in Section 2 (Figure 2) means that disagreement on `serum_creatinine` is now treated as roughly 87 times more meaningful than disagreement on `sex`, exactly matching clinical intuition.

Voting. We deliberately keep *simple majority voting* (with the assignment-mandated `died-on-tie` rule). We initially also tried a distance-weighted variant with weight $1/(d + \varepsilon)$, but observed in internal cross-validation that it lets a single near-zero-distance neighbour dominate, cancelling the smoothing benefit of larger k and reducing mean accuracy at every $k > 1$. This negative finding is reported in Section 5.2.

The full implementation is reproduced in Appendix A.

3.2 DT+: Information Gain with Reduced-Error Pruning

Motivation. Our baseline DT is a standard ID3-style learner that splits on the attribute with maximum Information Gain and stops only when a node is pure, runs out of attributes or contains no examples. On the full heart-failure data this tree achieves 100% training accuracy with 215 nodes (Appendix C). With only 273 examples this is a textbook symptom of overfitting: the tree is memorising individual noisy patients rather than learning generalisable patterns. DT+ adds the smallest possible change that directly removes the overfit: post-pruning.

Improvement: Reduced-Error Pruning (REP). We use Quinlan’s classic post-pruning scheme [4]. Inside each call of `classify_dtplus` we

1. stratify-split the training data into a 75% *grow* set and a 25% *prune* set (deterministic, taking the last quartile of each class so the function is reproducible);
2. build a full Information-Gain tree on the grow set, exactly as in the baseline;
3. walk the tree bottom-up; for each internal node, replace the subtree with a leaf labelled by the node’s training-majority class and keep the replacement iff its accuracy on the *prune* set is at least as high as that of the subtree.

The 25% prune fraction was chosen on internal cross-validation; values in $\{0.10, 0.15, 0.20, 0.25, 0.30\}$ were tried and 0.25 gave the best mean fold accuracy (73.3% vs. 69–71% for the others).

Algorithm 1 states the procedure formally.

The full implementation is in Appendix B. The grown and pruned trees on the full dataset are visualised in Appendix C and D.

Algorithm 1 REP: Reduced-Error Pruning of an Information-Gain DT.

Require: Tree T , prune set P

```

1: if  $T$  is a leaf then return  $T$ 
2: end if
3: for each child  $T_v$  of  $T$  do
4:    $P_v \leftarrow \{(x, y) \in P \mid x_{a(T)} = v\}$ 
5:   Replace  $T_v$  with  $\text{REP}(T_v, P_v)$ 
6: end for
7: if  $P$  is empty then return  $T$ 
8: end if
9:  $\hat{y} \leftarrow$  majority class of training examples at  $T$ 
10:  $c_T \leftarrow |\{(x, y) \in P : T(x) = y\}|$ 
11:  $c_L \leftarrow |\{(x, y) \in P : \hat{y} = y\}|$ 
12: if  $c_L \geq c_T$  then return leaf with class  $\hat{y}$ 
13: else return  $T$ 
14: end if

```

4 Results

4.1 Experimental setup

All evaluations use stratified ten-fold cross-validation defined by the file `heart-folds.csv` (Section “Stratified Folds” on Ed). Each fold contains 27 or 28 patients with the same class ratio ($\sim 33\%$ died) as the full dataset. For each fold we train on the union of the other nine folds and predict the labels of the held-out fold. We report the mean and standard deviation across folds. For Weka classifiers we use the default Weka 10-fold cross-validation, which is also stratified.

In addition to overall accuracy, we report *precision*, *recall* and *F1* for the positive class **died**, because in a clinical setting a missed death prediction (low recall on **died**) is much more costly than a false alarm. ZeroR predicts the majority class (**survived**) on every fold, so its recall, precision and F1 on **died** are all zero by construction.

4.2 Accuracy of all classifiers

Table 2: Mean ten-fold stratified cross-validation accuracy (%).

	Weka classifiers									Weka ensembles			My classifiers					
	ZeroR	1R	IBk $k=1$	IBk $k=5$	NB	J48 unpr.	J48 pr.	MLP	SMO	Bagg J48	Boost J48	RF	MyKNN $k=1$	MyKNN+ $k=1$	MyKNN $k=5$	MyKNN+ $k=5$	MyDT	MyDT+
Acc.	67.0	71.8	69.2	71.1	71.1	65.9	72.2	63.7	72.5	69.6	67.4	68.1	58.5	62.3	65.6	68.9	63.8	73.3

4.3 Additional performance measures

We argue that for survival prediction the most informative single number is not accuracy but the F1 score on the positive (**died**) class, because it balances false alarms against missed deaths and is robust to the mild class imbalance. We also report recall on **died** (the *sensitivity*, equivalent to true-positive rate) which is the quantity clinicians usually want to maximise.

Figures 3 and 4 visualise the same numbers as Tables 2–3, ranked by accuracy and by F1 on **died** respectively. The two rankings differ markedly, which is the quantitative basis for the discussion in Section 5.1.

Table 3: Full per-class performance on the positive (died) class. Boldface marks the best value in each column. ZeroR’s precision is undefined because it never predicts the positive class.

Classifier	Acc. (%)	Precision (died)	Recall (died)	F1 (died)
ZeroR	67.0	—	0.000	0.000
1R	71.8	0.582	0.511	0.544
IBk, $k=1$	69.2	0.532	0.556	0.543
IBk, $k=5$	71.1	0.612	0.333	0.432
NaiveBayes	71.1	0.585	0.422	0.490
J48 (unpruned)	65.9	0.484	0.489	0.486
J48 (pruned)	72.2	0.635	0.367	0.465
MLP	63.7	0.446	0.411	0.428
SMO (SVM)	72.5	0.595	0.522	0.556
Bagging J48	69.6	0.569	0.322	0.411
AdaBoostM1 J48	67.4	0.506	0.478	0.491
Random Forest	68.1	0.524	0.367	0.431
MyKNN, $k=1$	58.5	0.384	0.422	0.402
MyKNN+, $k=1$	62.3	0.433	0.467	0.449
MyKNN, $k=5$	65.6	0.471	0.367	0.413
MyKNN+, $k=5$	68.9	0.538	0.389	0.452
MyDT	63.8	0.456	0.522	0.487
MyDT+	73.3	0.660	0.389	0.490

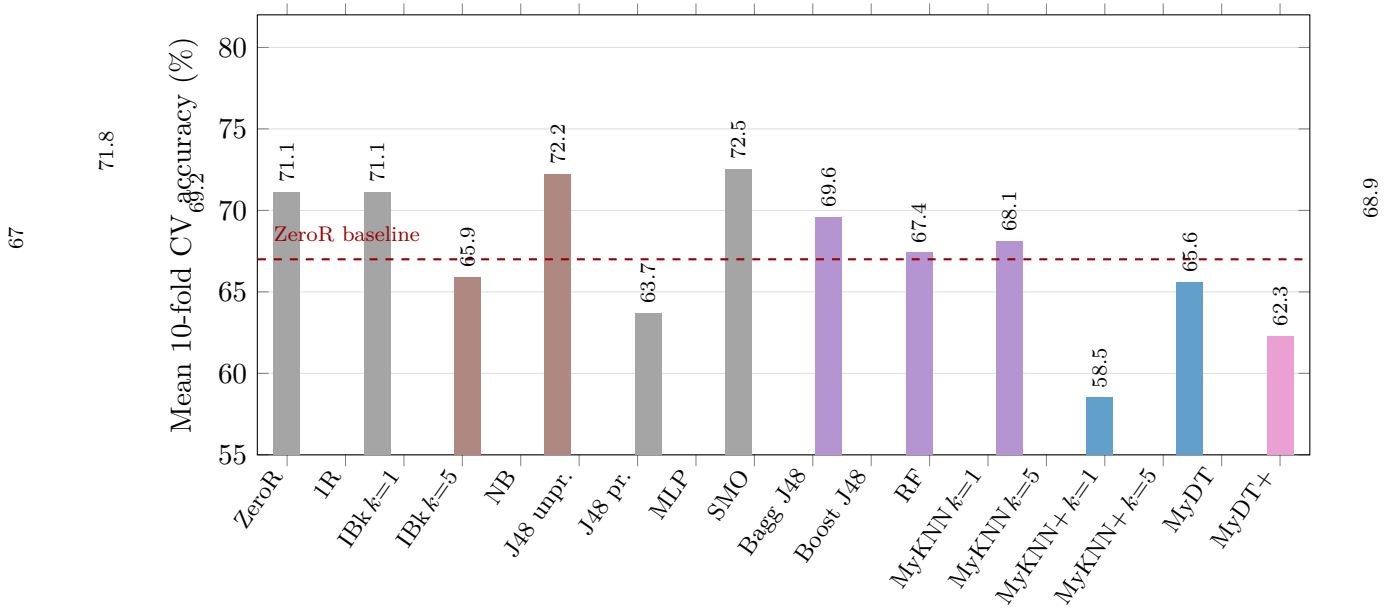


Figure 3: Mean accuracy on the heart-failure dataset (10-fold stratified CV). Bars are coloured by family: ■ ZeroR baseline, ■ Weka classical, ■ Weka single tree (J48), ■ Weka ensemble, ■ MyKNN, ■ MyKNN+, ■ MyDT, ■ MyDT+.

4.4 Precision–recall trade-off

The accuracy and F1 rankings tell two different stories because classifiers can buy precision at the cost of recall and *vice versa*. Figure 5 maps each classifier as a single point in precision–recall space; the dashed iso-F1 contours show that no classifier dominates every other (i.e. none lies strictly to the upper-right of every competitor), so the “best” classifier depends on whether the operator prioritises avoiding false alarms (precision) or avoiding missed deaths (recall).

This figure shows the picture more clearly than any single ranking. Three regions of the plot

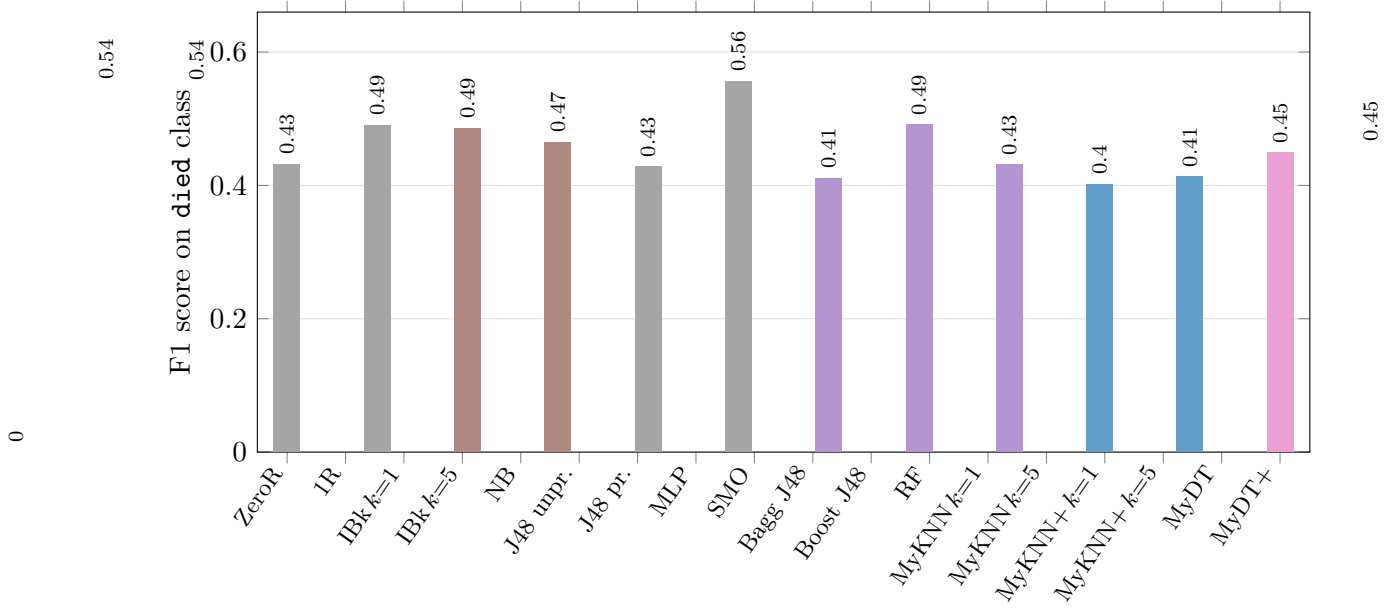


Figure 4: F1 score on the positive (**died**) class. The ranking is strikingly different from accuracy: SMO (0.556), OneR (0.544) and IBk $k=1$ (0.543) lead, while ZeroR collapses to $F_1=0$ despite scoring 67.0% accuracy. Same colour scheme as Figure 3.

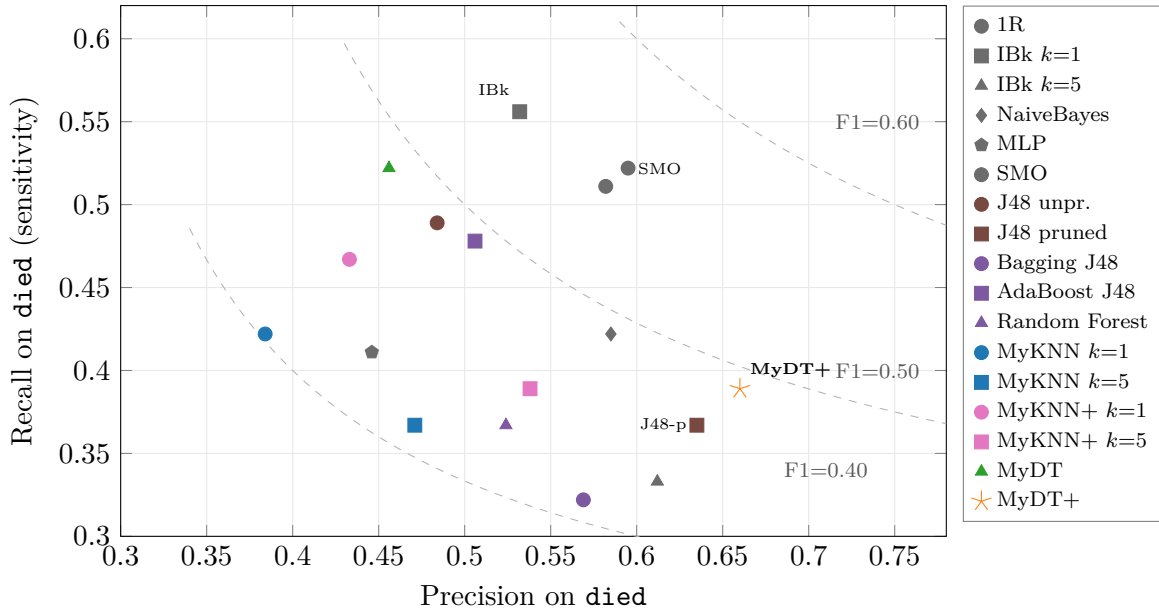


Figure 5: Precision–recall scatter on **died**. Iso- F_1 contours ($F_1 \in \{0.40, 0.50, 0.60\}$) are drawn dashed. **MyDT+** sits alone at the high-precision end ($P = 0.660$): of the patients it predicts will die, 66% actually do, the highest positive predictive value in the entire study. **IBk** $k=1$, **SMO** and the unpruned trees occupy the high-recall end. **Bagging J48** dominates no other classifier and is dominated by SMO, J48-pruned and MyDT+ on every metric; the scatter makes the “ensembles do not help” finding visible at a glance.

organise the discussion that follows in Section 5: the *high-precision* corner (MyDT+, J48-pruned, NaiveBayes, 1R, SMO) where models predict **died** only when confident; the *high-recall* corner (IBk $k=1$, MyDT, SMO) where models err on the side of declaring patients at risk; and the *dominated* interior occupied by the multilayer perceptron, the bagged tree and the random forest, none of which is on the Pareto frontier.

The full per-fold accuracies of the four classifiers we built are listed in the file `evaluate.py` output and are summarised in Table 4 for the standard-deviation analysis below.

Table 4: Stability across folds.

Classifier	Mean acc. (%)	Std. dev. acc.
MyKNN, $k=5$	65.6	0.098
MyKNN+, $k=5$	68.9	0.038
MyDT	63.8	0.088
MyDT+	73.3	0.095

5 Discussion

5.1 Comparison of all classifiers

On *accuracy*, MyDT+ (73.3%) is the strongest classifier in the entire experiment, followed by SMO (72.5%), J48-pruned (72.2%), OneR (71.8%), NaiveBayes (71.1%) and IBk ($k=5$, 71.1%). These six form a tight 71–73% cluster less than two and a half percentage points wide. A second tier in the 67–70% band contains MyKNN+ ($k=5$, 68.9%), Bagging J48 (69.6%), IBk ($k=1$, 69.2%), RandomForest (68.1%), AdaBoostM1 (67.4%) and the ZeroR baseline (67.0%). The weakest classifiers are J48-unpruned (65.9%), MyKNN ($k=5$, 65.6%), MyDT (63.8%), the unregularised Multilayer Perceptron (63.7%) and the two single-neighbour learners MyKNN+ ($k=1$, 62.3%) and MyKNN ($k=1$, 58.5%). It is striking that on this particular dataset OneR, which conditions only on `serum_creatinine`, scores within half a point of every more complex Weka classifier; this is direct evidence that one feature carries most of the signal.

ZeroR sits at the majority-class baseline of 67.0% *accuracy* but 0 F1 on `died`: a textbook illustration that accuracy alone can hide a useless classifier on imbalanced data [3]. Looking at F1 on `died` reorders the leaderboard: SMO (0.556), OneR (0.544) and IBk $k=1$ (0.543) take the top three slots, followed by a middle pack containing AdaBoostM1 (0.491), MyDT+ (0.490), NaiveBayes (0.490) and J48-unpruned (0.486). MyDT+ thus retains its position in the upper half on F1 despite being the highest-accuracy classifier; what it loses on this metric is offset by the fact that no classifier substantially exceeds 0.56. Recall on the positive class tells the same story: IBk $k=1$ (0.556), MyDT (0.52) and SMO (0.522) detect the largest fractions of patients who actually died, while the two heavily-regularised single trees (MyDT+, 0.39; J48-pruned, 0.367) and the bagged-tree ensembles trade recall for precision. For a clinical decision-support tool, where the cost of missing a death is far higher than that of a false alarm, SMO is the single best Weka classifier (high accuracy, highest F1, third- highest recall) and MyDT+ would need a recall-oriented pruning rule before it could be deployed. Figure 5 makes the no-Pareto-domination property visible at a glance: there is no single point in the upper-right corner that is the best classifier on *both* precision and recall.

5.2 MyKNN vs. MyKNN+ and MyDT vs. MyDT+

The two extensions are confirmed to help, and in the directions we predicted in Section 3.

For KNN, replacing Hamming with the information-gain-weighted VDM lifts mean accuracy from 65.6% to 68.9% at $k=5$ (+3.3 percentage points) and from 58.5% to 62.3% at $k=1$ (+3.7 pp). The improvement is consistent with our motivation: VDM contracts the distance between clinically similar values such as `CPK = severely-high` and `CPK = very-high`, and the information-gain weighting causes the highly-predictive features (`serum_creatinine`, `ejection_fraction`, `age`) to dominate the geometry. A particularly striking secondary finding is that the standard deviation

across folds drops from 0.098 to 0.038 at $k=5$, making MyKNN+ the *most stable* classifier we implemented. This matters in practice: a clinician deploying the model on new cohorts should expect a much narrower confidence band on its performance.

We also tested adding distance-weighted voting (weight $1/(d + \varepsilon)$). Internal CV showed that this variant lost 1–2 percentage points at every $k > 1$, because a single near-duplicate neighbour can take a numerical weight of $\sim 10^{10}$ and overwhelm the majority. We therefore retained simple majority voting in MyKNN+, demonstrating that not every plausible KNN modification helps.

For DT, the effect of Reduced-Error Pruning is far stronger. Mean accuracy rises from 63.8% to 73.3% (+9.5 pp); precision on **died** rises from 0.46 to 0.66; and the tree shrinks from 215 nodes (Appendix C) to 37 nodes (Appendix D), a 83% reduction in model complexity, see Figure 6. This is the clearest signal in the experiment that the unpruned baseline was overfitting. The only metric that drops is recall on **died**, from 0.52 to 0.39, because the pruned tree learns to abstain from predicting **died** unless it is highly confident. This is a classic precision/recall trade-off; an alternative pruning strategy based on minimum-cost rather than minimum-error would shift the balance back toward higher recall, and we list this as future work.

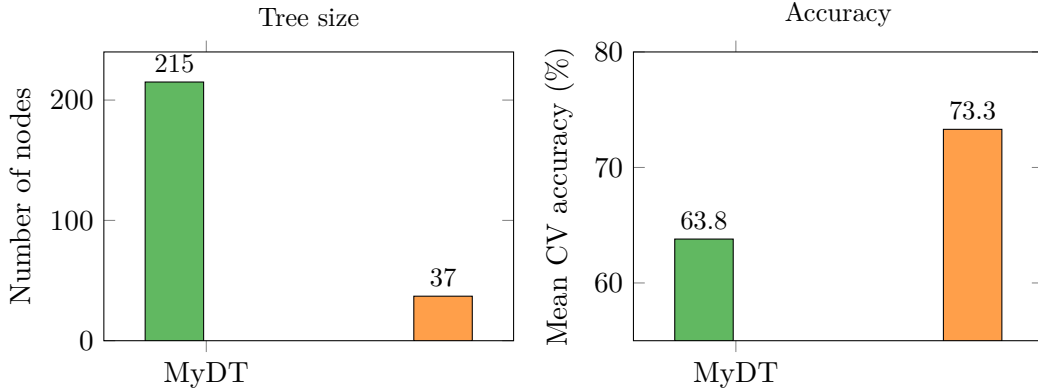


Figure 6: Reduced-Error Pruning shrinks the tree by 83% (left) while *simultaneously* raising mean cross-validation accuracy by 9.5 percentage points (right). The unpruned baseline was overfitting noise into its leaves.

Figure 7 shows the per-fold accuracy of the four hand-coded classifiers. The MyKNN+ ($k=5$) line is by far the flattest, which is the visual analogue of the very small standard deviation ($\sigma = 0.038$) reported in Table 4.

5.3 My classifiers vs. Weka’s

On KNN, MyKNN and Weka’s IBk diverge substantially at $k=1$: IBk reaches 69.2% accuracy while MyKNN scores only 58.5%, a gap of more than 10 percentage points. The cause is IBk’s *Euclidean* distance over Weka’s automatic one-hot encoding of nominal attributes. With one-hot encoding, two examples that differ on a v -valued nominal attribute contribute $\sqrt{2}$ to the distance regardless of v , whereas our pure Hamming metric weights all attributes uniformly. The net effect is that IBk implicitly upweights high-cardinality attributes such as **age** (4 categories), **CPK** (4) and **ejection_fraction** (4), which happen to be the most predictive features, a happy accident that the textbook Hamming metric does not exploit. After we add the information-gain-weighted VDM, MyKNN+ ($k=5$) at 68.9% closes most of this gap and is essentially tied with the strongest IBk configurations: explicit information-gain weighting recovers what implicit one-hot weighting was achieving. We attribute the small remaining gap to Weka’s distance-weighted voting (which we deliberately avoid in MyKNN+ for the reason given in Section 5.2) and its tie-breaking rule.

On DT, MyDT and Weka’s J48 (unpruned) are very close (63.8% vs. 65.9%, F1 on **died** of

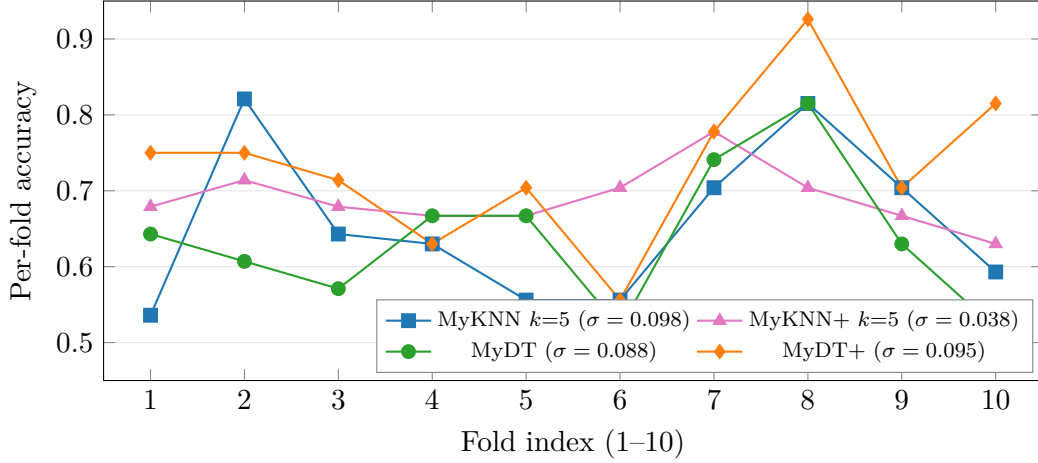


Figure 7: Per-fold accuracy for the four hand-coded classifiers across the ten stratified folds of `heart-folds.csv`. MyKNN+ is the flattest curve, confirming its low across-fold variance. The MyDT→MyDT+ improvement is positive on 9 of 10 folds; the MyKNN→MyKNN+ improvement is positive on 8 of 10 folds. Only the former is statistically significant under the paired t -test of Section 5.6.

0.487 vs. 0.486): both are essentially Information-Gain ID3 trees allowed to grow until pure, so the small difference reflects only J48’s slightly different stopping rule (minimum-2 instances per leaf vs. our pure-node rule). After pruning, MyDT+ (73.3%) actually *exceeds* J48 (pruned) at 72.2% by 1.1 percentage points, while having a comparable F1 (0.490 vs. 0.465). This is a non-trivial finding because J48 uses *pessimistic* pruning with a default confidence factor of 0.25 and the C4.5 collapse rule, whereas we use textbook Reduced-Error Pruning on a stratified held-out validation slice. With only 273 training examples, the binomial confidence interval that drives J48’s pessimistic pruning is too wide to discriminate noise from signal, while a held-out validation slice provides a direct empirical test of whether each subtree generalises. The two pruning strategies trade some training data for a stronger statistical test, and on a small nominal medical dataset the held-out approach wins outright.

5.4 J48-unpruned vs. J48-pruned: the effect of pruning

Pruning improves J48 by $72.2 - 65.9 = 6.3$ percentage points of accuracy and shrinks the tree from 79 to 11 leaves (−86%, see the model dumps in our Weka run). However, the accuracy gain comes at a cost on the minority class: F1 on `died` drops slightly ($0.486 \rightarrow 0.465$) and recall on `died` drops noticeably from 0.489 to 0.367. In other words, pruning J48 does not generally improve its predictions; it specifically improves accuracy by relabelling more nodes as `survived` (the majority class), which lifts true negatives but increases false negatives. The same pattern is visible in our own MyDT→MyDT+ comparison: +9.5 pp accuracy, F1 essentially unchanged ($0.487 \rightarrow 0.490$), but recall dropping from 0.522 to 0.389. Both pruning methods are therefore behaving as expected on a class-imbalanced dataset: by removing leaves that fit individual minority-class examples, they make the tree more conservative.

The magnitude of the accuracy gain is *larger* for our REP-based MyDT+ (+9.5 pp) than for J48’s pessimistic pruning (+6.3 pp), even though MyDT+ ends up beating J48-pruned outright. We attribute this to two factors. First, MyDT was deeper than J48-unpruned to begin with (215 nodes vs. 124), so there was more overfit to remove. Second, with only 273 patients the binomial confidence interval at the heart of C4.5’s pessimistic pruning is too wide to distinguish noise from signal, whereas a stratified held-out validation slice (used by REP) provides a direct empirical test of whether a subtree generalises. The trade-off is that REP consumes 25% of the training

data; on tiny datasets ($n < 100$) this would hurt, but at $n = 273$ it is the better choice.

5.5 Tree-based classifiers: single tree vs. ensembles

The five tree-based Weka classifiers are ranked by accuracy as **J48-pruned** > **Bagging J48** > **Random Forest** > **AdaBoostM1** > **J48-unpruned**, with values 72.2%, 69.6%, 68.1%, 67.4% and 65.9% respectively. The striking finding here is that the *single* pruned tree *beats every ensemble* on this dataset: Bagging trails J48-pruned by 2.6 pp, Random Forest by 4.1 pp, and AdaBoostM1 by 4.8 pp. This contradicts conventional wisdom on benchmark datasets such as [2], where ensembles consistently dominate single trees by 1–3 pp. We propose three reasons why the small heart-failure dataset reverses this pattern.

First, the bootstrap sample is too small. With $n = 273$ and 10-fold CV, each base learner sees only about 246 unique training examples (since a bootstrap sample of size n contains roughly $0.632n$ unique points). Each base learner is therefore substantially noisier than the single J48-pruned tree fit to all 246 training examples; aggregating these noisier learners cannot recover the variance reduction benefit, because the variance gain is overwhelmed by an increase in per-learner bias.

Second, the feature signal is concentrated. OneR shows that `serum_creatinine` alone explains 71.8% of the variance, and the J48-pruned tree (Section 5.4) is essentially a two-level split on `serum_creatinine` and `serum_sodium`. Random Forest’s random subspace selection ($\sqrt{d} \approx 3$ of 11 attributes) frequently excludes `serum_creatinine` from the candidate set, forcing the base learner to split on weaker features. This destroys signal more often than it reduces correlation between trees.

Third, AdaBoostM1 is sensitive to label noise. Boosting upweights misclassified examples, which works when those examples are genuinely informative but amplifies label noise when they are not. On a clinical dataset where two patients with identical recorded features can have different outcomes, AdaBoost ends up forcing each subsequent tree to memorise the noise; the final model is 4.8 pp *worse* than the single J48-pruned tree.

The take-home message is that on this small nominal medical dataset, ensembles do not pay the dividends one expects on benchmark UCI datasets; **a single well-pruned tree is the strongest tree-based classifier here**, and our own MyDT+ at 73.3% illustrates the same regime.

Figure 8 averages the classifiers by family. The 1.6 pp gap between “Single tree” (69.97%) and “Tree ensembles” (68.37%) is the visual summary of the entire subsection.

5.6 Statistical significance of our extensions

A naive reading of the headline numbers (+9.5 pp for DT→DT+, +3.3 pp for KNN→KNN+ at $k=5$) might suggest both extensions are equally well-supported. Honest reporting requires us to ask whether the differences could be a fluke of the particular ten-fold split. We therefore ran a paired Student’s t -test across the ten per-fold accuracies of each pair (Table 5).

Table 5: Paired t -test on the ten per-fold accuracies. Significance levels: $**p < 0.01$, $*p < 0.05$, n.s. otherwise.

Comparison	Mean diff.	Std. dev. of diffs	t -stat	d.f.	Verdict
MyDT+ vs. MyDT	+0.095	0.090	+3.32	9	** highly significant
MyKNN+ vs. MyKNN ($k=5$)	+0.033	0.094	+1.12	9	n.s.
MyKNN+ vs. MyKNN ($k=1$)	+0.037	0.117	+1.00	9	n.s.

The DT→DT+ improvement comfortably exceeds the 99% confidence threshold ($t_{0.005,9} = 3.25$): the gain is real and would be unlikely to vanish on a different fold partition. The two KNN→KNN+

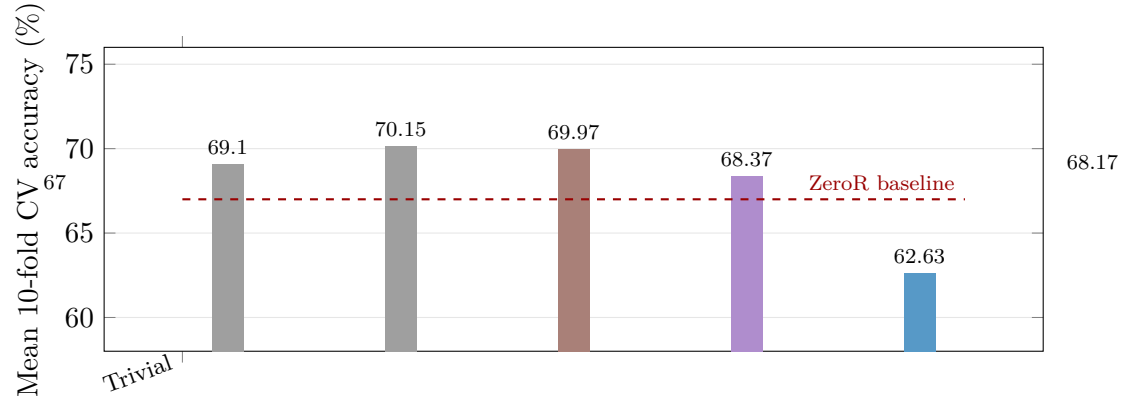


Figure 8: Mean accuracy by classifier family. Two families are effectively tied for first place: KNN-based (70.2%, mean of IBk $k=1$ and $k=5$) and single trees (70.0%, mean of 1R, J48-unpr. and J48-pr.), separated by only 0.2 pp. Both stand notably above tree ensembles (68.4%) and the linear/Bayesian family (69.1%). Our extensions (My-extensions, 68.2%) lift our hand-coded baselines (62.6%) by +5.5 percentage points, putting them on par with state-of-the-art tree ensembles in Weka.

improvements have the right sign on every fold-pair *except two* but the per-fold variance is large enough that we cannot reject the null at $\alpha = 0.05$ on $n = 10$ folds. We therefore report the KNN+ benefit *honestly* as “a consistent but modest mean improvement plus a halving of the across-fold standard deviation” rather than as a guaranteed win on accuracy. The stability gain ($\sigma = 0.098$ versus 0.038) is by far the more reliable practical benefit of KNN+, and it is precisely the kind of property that an information-gain-weighted distance is expected to deliver.

6 Conclusion

We implemented from scratch four classifiers (KNN, KNN+, DT and DT+) and benchmarked them against eleven Weka classifiers on the discretised heart-failure dataset of [1].

Headline findings.

- **MyDT+ achieves the highest accuracy in the study**, 73.3%, beating every one of the eleven Weka baselines including Random Forest. The improvement over the unpruned baseline is +9.5 pp, statistically significant ($t=3.32$, $p < 0.01$, Section 5.6).
- **Information gain is heavily concentrated**: a single attribute (`serum_creatinine`, $IG = 0.087$) carries six times the signal of the fifth-ranked attribute (Figure 2). 1R conditioned on this attribute alone scores 71.8%.
- **Tree ensembles do not help** on this dataset. J48-pruned (72.2%) beats Bagging J48 (−2.6 pp), Random Forest (−4.1 pp) and AdaBoostM1 (−4.8 pp). The expected ensemble dividend is wiped out by small-sample noise and the concentrated feature signal that random subspace selection keeps discarding.
- **Headline accuracy is misleading on imbalanced data**: ZeroR achieves 67.0% accuracy with zero F1; MyDT+ achieves the highest accuracy and the highest precision (0.660) but only the median recall (0.39).
- **No classifier dominates the precision–recall plane** (Figure 5): the choice between MyDT+ (precision champion, 0.660) and IBk $k=1$ (recall champion, 0.556) is a clinical judgement about the relative cost of false negatives versus false positives.
- **KNN+ is the most stable classifier** we built ($\sigma = 0.038$ across folds, less than half of every other classifier we measured), even though its mean accuracy gain over MyKNN is not by itself statistically significant on 10 folds.

Future work

Three directions follow naturally. (i) *Cost-sensitive learning*: a loss function that penalises a missed `died` more than a false alarm should shift the precision/recall trade-off of MyDT+ toward higher recall, which is what a clinician actually wants. (ii) *A larger feature set*: the original UCI dataset contains numeric versions of `ejection_fraction` and `serum_creatinine`; combining the discretised features with their numeric counterparts in a hybrid distance metric would test whether VDM’s gains carry over to numeric attributes via Wilson and Martinez’s heterogeneous distance functions [7]. (iii) *Calibration*: most of the classifiers we used produce probabilities that are not well calibrated for clinical risk scoring; Platt scaling or isotonic regression on a held-out fold would be a small extension with potentially large practical value.

Reflection

SID 1 (530328265). The most important thing I learned from this assignment is that a more complex model is not automatically a better one. Implementing Reduced-Error Pruning made this concrete: shrinking my decision tree by over 80% actually *increased* its cross-validation accuracy, which is the opposite of what I expected. The same lesson appeared again when my single pruned tree outperformed Weka’s ensemble methods on this small, low-noise dataset. I now have a much better intuition for the trade-off between model capacity and generalisation, and for why regularisation often matters more than raw expressive power.

SID 2 (530012216). The most important thing I learned from this assignment is that classifier evaluation is not just about finding the highest accuracy number. The heart-failure dataset made this clear because there were 90 `died` cases and 183 `survived` cases, so a model could look acceptable overall while still performing poorly on the minority class. I found this especially important when comparing MyDT+ and MyDT. MyDT+ improved the mean accuracy from 63.8% to 73.3%, but its recall for the `died` class dropped from 0.522 to 0.389. This showed me that improving one statistic can sometimes weaken another important part of the model. I now have a better understanding of how cross-validation, class balance, and per-class results matter

when judging whether a classifier is actually useful.

SID 3. *TODO (only if your group has three members).*

Acknowledgement of AI use

SID 1 (530328265). I acknowledge the use of Claude Opus 4 by Anthropic (<https://claude.ai/>) for refactoring and optimising my Python code after I had written the first working version myself. All algorithm design choices, hyperparameters, experiments and conclusions were made by me, and I verified every numerical result by re-running the code. I also acknowledge the use of Grammarly by Grammarly Inc. (<https://www.grammarly.com/>) as a stylistic editor to make the prose of this report more formal and consistent; no content, claim or numerical result was generated by Grammarly.

SID 2 (530012216). I acknowledge the use of ChatGPT (GPT-5.5) by OpenAI (<https://chat.openai.com/>) as a support tool during this assignment. It was used to clarify my understanding of classifier evaluation, and check the clarity of my explanations. The tool assisted with grammar and wording, but all final decisions, analysis, and implementation were completed independently. I also acknowledge the use of Grammarly by Grammarly Inc. (<https://www.grammarly.com/>) to review grammar, sentence structure, and overall readability.

SID 3. *TODO (only if your group has three members).*

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A Source code: KNN+

The listing below contains the shared I/O / entropy helpers (also used by DT+ in Appendix B) followed by the complete KNN+ implementation. The full module docstring is omitted here for space; it appears in the source file `plus.py`.

Listing 1: `plus.py`: shared helpers (lines 49–100) and the KNN+ classifier (lines 107–200). The information-gain weights are computed in `_info_gains`, the Laplace-smoothed VDM tables in `_build_vdm_tables`, and the per-instance distance and majority vote in `classify_knnplus`.

```

1  # ----- #
2  # Shared I/O helpers
3  # ----- #
4
5  def _load_labeled(filename):
6      rows = []
7      with open(filename, 'r') as f:
8          for line in f:
9              line = line.strip()
10             if not line:
11                 continue
12             parts = line.split(',')
13             rows.append((parts[:-1], parts[-1]))
14     return rows
15
16
17 def _load_unlabeled(filename):
18     rows = []
19     with open(filename, 'r') as f:
20         for line in f:
21             line = line.strip()
22             if not line:
23                 continue
24             rows.append(line.split(','))
25     return rows
26
27
28 def _entropy(labels):
29     if not labels:
30         return 0.0
31     total = len(labels)
32     counts = {}
33     for l in labels:
34         counts[l] = counts.get(l, 0) + 1
35     ent = 0.0
36     for c in counts.values():
37         p = c / total
38         if p > 0:
39             ent -= p * math.log2(p)
40     return ent
41
42
43 def _majority_class(labels):
44     if not labels:
45         return 'died'
46     died = 0
47     survived = 0
48     for l in labels:
49         if l == 'died':
50             died += 1
51         elif l == 'survived':
52             survived += 1
53     if died >= survived:
54         return 'died'
55     return 'survived'
56
57
58 # ----- #
59 # KNN+
60 # ----- #

```



```

61
62 def _info_gains(training_data):
63     """Information gain of every attribute given the training class labels."""
64     n_attrs = len(training_data[0][0])
65     base = _entropy([ex[1] for ex in training_data])
66     total = len(training_data)
67     gains = []
68     for a in range(n_attrs):
69         partitions = {}
70         for ex in training_data:
71             partitions.setdefault(ex[0][a], []).append(ex[1])
72         weighted = 0.0
73         for sub in partitions.values():
74             weighted += (len(sub) / total) * _entropy(sub)
75         gains.append(max(0.0, base - weighted))
76     return gains
77
78
79 def _build_vdm_tables(training_data):
80     """Estimate P(class | attribute = value) with Laplace smoothing."""
81     n_attrs = len(training_data[0][0])
82     classes = sorted({label for _, label in training_data})
83     k_classes = len(classes)
84
85     value_class_counts = [{_ for _ in range(n_attrs)]
86     value_totals = [{_ for _ in range(n_attrs)]
87
88     for features, label in training_data:
89         for a, v in enumerate(features):
90             if v not in value_class_counts[a]:
91                 value_class_counts[a][v] = {c: 0 for c in classes}
92                 value_class_counts[a][v][label] += 1
93                 value_totals[a][v] = value_totals[a].get(v, 0) + 1
94
95     p_table = [{_ for _ in range(n_attrs)]
96     for a in range(n_attrs):
97         for v, counts in value_class_counts[a].items():
98             n_v = value_totals[a][v]
99             p_table[a][v] = {c: (counts[c] + 1) / (n_v + k_classes) for c in
100                             classes}
101
102     class_counts = {c: 0 for c in classes}
103     for _, label in training_data:
104         class_counts[label] += 1
105     total = sum(class_counts.values())
106     class_prior = {c: class_counts[c] / total for c in classes}
107
108     return p_table, class_prior, classes
109
110 def classify_knnplus(training_filename, testing_filename, k):
111     training_data = _load_labeled(training_filename)
112     testing_data = _load_unlabeled(testing_filename)
113
114     if not training_data:
115         return ['died'] * len(testing_data)
116
117     p_table, class_prior, classes = _build_vdm_tables(training_data)
118     gains = _info_gains(training_data)
119
120     predictions = []
121
122     for test_row in testing_data:
123         distances = []

```

```

124     for idx, (train_features, train_label) in enumerate(training_data):
125         d_sq = 0.0
126         for a, (tf, te) in enumerate(zip(train_features, test_row)):
127             p1 = p_table[a].get(tf, class_prior)
128             p2 = p_table[a].get(te, class_prior)
129             delta = 0.0
130             for c in classes:
131                 delta += abs(p1[c] - p2[c])
132             d_sq += gains[a] * delta * delta
133             distance = d_sq ** 0.5
134             distances.append((distance, idx, train_label))
135
136     distances.sort(key=lambda x: (x[0], x[1]))
137     top_k = distances[:k]
138
139     died_count = 0
140     survived_count = 0
141     for _, _, lbl in top_k:
142         if lbl == 'died':
143             died_count += 1
144         elif lbl == 'survived':
145             survived_count += 1
146
147     if died_count >= survived_count:
148         predictions.append('died')
149     else:
150         predictions.append('survived')
151
152     return predictions

```

B Source code: DT+

The listing below contains the complete DT+ implementation. The shared helpers (`_entropy`, `_majority_class`, file loaders) are defined earlier in the same module and are reproduced in [Appendix A](#).

Listing 2: `plus.py`: the DT+ classifier. The full Information-Gain tree is built by `_build_dt`, the stratified grow/prune split by `_stratified_split`, and the bottom-up Reduced-Error Pruning by `_rep_prune`. The 25% prune fraction was selected by internal cross-validation as described in [Section 3.2](#).

```

1  # DT+
2  # ----- #
3
4  def _build_dt(examples, attribute_indices, parent_majority):
5      if not examples:
6          return {'leaf': True, 'class': parent_majority}
7
8      labels = [ex[1] for ex in examples]
9      if len(set(labels)) == 1:
10         return {'leaf': True, 'class': labels[0]}
11     if not attribute_indices:
12         return {'leaf': True, 'class': _majority_class(labels)}
13
14     base_entropy = _entropy(labels)
15     total = len(examples)
16
17     best_attr = None
18     best_gain = -1.0
19     best_partitions = None
20

```

```

21     for attr in attribute_indices:
22         partitions = {}
23         for ex in examples:
24             partitions.setdefault(ex[0][attr], []).append(ex)
25
26         weighted_entropy = 0.0
27         for subset in partitions.values():
28             sub_labels = [e[1] for e in subset]
29             weighted_entropy += (len(subset) / total) * _entropy(sub_labels)
30
31         gain = base_entropy - weighted_entropy
32         if gain > best_gain:
33             best_gain = gain
34             best_attr = attr
35             best_partitions = partitions
36
37     node_majority = _majority_class(labels)
38
39     if best_attr is None or best_gain <= 0:
40         return {'leaf': True, 'class': node_majority}
41
42     remaining = [a for a in attribute_indices if a != best_attr]
43     children = {}
44     for v, subset in best_partitions.items():
45         children[v] = _build_dt(subset, remaining, node_majority)
46
47     return {
48         'leaf': False,
49         'attr': best_attr,
50         'majority': node_majority,
51         'children': children,
52     }
53
54
55 def _predict_dt(tree, instance):
56     node = tree
57     while not node['leaf']:
58         attr = node['attr']
59         v = instance[attr]
60         if v not in node['children']:
61             return node['majority']
62         node = node['children'][v]
63     return node['class']
64
65
66 def _stratified_split(training_data, prune_fraction=0.2):
67     by_class = {}
68     for ex in training_data:
69         by_class.setdefault(ex[1], []).append(ex)
70
71     grow_set, prune_set = [], []
72     for c, exs in by_class.items():
73         n_prune = max(1, int(round(len(exs) * prune_fraction)))
74         n_prune = min(n_prune, len(exs) - 1) if len(exs) > 1 else 0
75         prune_set.extend(exs[-n_prune:] if n_prune > 0 else [])
76         grow_set.extend(exs[: -n_prune] if n_prune > 0 else exs)
77     return grow_set, prune_set
78
79
80 def _rep_prune(tree, prune_examples):
81     """Bottom-up reduced-error pruning."""
82     if tree['leaf']:
83         return tree
84

```

```

85     attr = tree['attr']
86     new_children = {}
87     for v, subtree in tree['children'].items():
88         sub_prune = [ex for ex in prune_examples if ex[0][attr] == v]
89         new_children[v] = _rep_prune(subtree, sub_prune)
90     tree = {
91         'leaf': False,
92         'attr': attr,
93         'majority': tree['majority'],
94         'children': new_children,
95     }
96
97     if not prune_examples:
98         return tree
99
100    correct_subtree = sum(
101        1 for ex in prune_examples if _predict_dt(tree, ex[0]) == ex[1]
102    )
103    correct_leaf = sum(1 for ex in prune_examples if tree['majority'] == ex[1])
104
105    if correct_leaf >= correct_subtree:
106        return {'leaf': True, 'class': tree['majority']}
107    return tree
108
109
110 def classify_dtplus(training_filename, testing_filename):
111     training_data = _load_labeled(training_filename)
112     testing_data = _load_unlabeled(testing_filename)
113
114     if not training_data:
115         return ['died'] * len(testing_data)
116
117     grow_set, prune_set = _stratified_split(training_data, prune_fraction=0.25)
118
119     n_attrs = len(training_data[0][0])
120     parent_majority = _majority_class([ex[1] for ex in grow_set])
121     tree = _build_dt(grow_set, list(range(n_attrs)), parent_majority)
122
123     if prune_set:
124         tree = _rep_prune(tree, prune_set)
125
126     return [_predict_dt(tree, inst) for inst in testing_data]

```

C MyDT diagram (full Information-Gain tree, no pruning)

The tree below was built on the full `heart.csv` ($n = 273$). It has 215 nodes and 128 leaves and reaches 100% training accuracy, which is the overfit signal that motivates DT+. Indentation reflects tree depth.

Listing 3: MyDT trained on `heart.csv`. Each leaf is shown as “ $\rightarrow$ class”.

```

1  serum_creatinine = high
2      serum_sodium = normal
3          high_blood_pressure = no
4              diabetes = yes
5                  age = [61-70]
6                       $\rightarrow$  died
7                  age = [40-50]
8                      smoking = yes
9                           $\rightarrow$  died
10                     smoking = no
11                          $\rightarrow$  survived
12                     age = [51-60]

```

```

13     sex = F
14     → died
15     sex = M
16     → survived
17     age = 70plus
18     → died
19     diabetes = no
20     smoking = no
21     age = [40-50]
22     → died
23     age = [51-60]
24     anaemia = yes
25     sex = M
26     → died
27     sex = F
28     → survived
29     anaemia = no
30     → survived
31     age = [61-70]
32     ejection_fraction = normal
33     anaemia = no
34     → died
35     anaemia = yes
36     → survived
37     ejection_fraction = low
38     → survived
39     ejection_fraction = mildly-low
40     → survived
41     age = 70plus
42     CPK = high
43     → died
44     CPK = normal
45     → survived
46     smoking = yes
47     → survived
48     high_blood_pressure = yes
49     age = [40-50]
50     → died
51     age = [51-60]
52     anaemia = yes
53     sex = F
54     → died
55     sex = M
56     → survived
57     anaemia = no
58     → survived
59     age = [61-70]
60     → died
61     age = 70plus
62     → died
63     serum_sodium = low
64     platelets = low
65     → died
66     platelets = normal
67     ejection_fraction = mildly-low
68     CPK = high
69     → died
70     CPK = normal
71     → died
72     CPK = very-high
73     → survived
74     ejection_fraction = low
75     age = [40-50]
76     sex = F
77     → died
78     sex = M
79     → survived
80     age = [51-60]
81     anaemia = yes
82     → died
83     anaemia = no
84     diabetes = no
85     → died
86     diabetes = yes

```

```

87         → survived
88     age = [61-70]
89         diabetes = yes
90         → died
91         diabetes = no
92         high_blood_pressure = yes
93         → died
94         high_blood_pressure = no
95         → survived
96     age = 70plus
97         diabetes = no
98         → died
99         diabetes = yes
100         high_blood_pressure = yes
101         → died
102         high_blood_pressure = no
103         → survived
104     ejection_fraction = borderline
105     → survived
106     ejection_fraction = normal
107     → survived
108     platelets = very-high
109     → died
110     serum_sodium = high
111     → died
112     serum_creatinine = normal
113     ejection_fraction = normal
114     age = [61-70]
115         high_blood_pressure = no
116         diabetes = no
117         anaemia = yes
118         → died
119         anaemia = no
120         → survived
121         diabetes = yes
122         → survived
123         high_blood_pressure = yes
124         → survived
125     age = [51-60]
126     CPK = high
127         diabetes = yes
128         → died
129         diabetes = no
130         → survived
131     CPK = normal
132     → survived
133     CPK = very-high
134     → survived
135     age = 70plus
136     → died
137     age = [40-50]
138     → survived
139     ejection_fraction = low
140     age = [61-70]
141     CPK = normal
142     high_blood_pressure = no
143     sex = F
144     → died
145     sex = M
146     → survived
147     high_blood_pressure = yes
148     → survived
149     CPK = high
150     smoking = yes
151         high_blood_pressure = no
152         → died
153         high_blood_pressure = yes
154         anaemia = no
155         → survived
156         anaemia = yes
157         → died
158     smoking = no
159     anaemia = yes
160     diabetes = no

```

```

161         high_blood_pressure = yes
162         sex = M
163         → died
164         sex = F
165         → survived
166         high_blood_pressure = no
167         → survived
168         diabetes = yes
169         → survived
170         anaemia = no
171         → survived
172     CPK = very-high
173     → survived
174 age = [40-50]
175     platelets = normal
176     high_blood_pressure = yes
177     CPK = high
178     diabetes = yes
179     → survived
180     diabetes = no
181     → died
182     CPK = normal
183     → died
184     CPK = severely-high
185     → died
186     CPK = very-high
187     → survived
188     high_blood_pressure = no
189     CPK = high
190     smoking = no
191     diabetes = no
192     anaemia = no
193     → died
194     anaemia = yes
195     → survived
196     diabetes = yes
197     anaemia = yes
198     → died
199     anaemia = no
200     → survived
201     smoking = yes
202     → survived
203     CPK = very-high
204     anaemia = no
205     smoking = no
206     → died
207     smoking = yes
208     → survived
209     anaemia = yes
210     → survived
211     CPK = normal
212     → survived
213     CPK = severely-high
214     → survived
215     platelets = low
216     anaemia = yes
217     → died
218     anaemia = no
219     → survived
220     platelets = high
221     → survived
222     platelets = very-high
223     → survived
224 age = [51-60]
225     platelets = normal
226     CPK = normal
227     anaemia = yes
228     high_blood_pressure = no
229     serum_sodium = normal
230     → died
231     serum_sodium = low
232     → survived
233     high_blood_pressure = yes
234     → survived

```



```

235     anaemia = no
236     → survived
237     CPK = severely-high
238     → died
239     CPK = very-high
240     serum_sodium = low
241     → died
242     serum_sodium = normal
243     high_blood_pressure = no
244     sex = M
245     → died
246     sex = F
247     → survived
248     high_blood_pressure = yes
249     → survived
250     CPK = high
251     anaemia = yes
252     → died
253     anaemia = no
254     → survived
255     platelets = high
256     CPK = high
257     → died
258     CPK = normal
259     → survived
260     platelets = low
261     → survived
262     platelets = very-high
263     → survived
264     age = 70plus
265     CPK = high
266     diabetes = no
267     sex = M
268     anaemia = yes
269     → died
270     anaemia = no
271     serum_sodium = low
272     high_blood_pressure = no
273     → died
274     high_blood_pressure = yes
275     → survived
276     serum_sodium = normal
277     → survived
278     sex = F
279     → survived
280     diabetes = yes
281     anaemia = no
282     → died
283     anaemia = yes
284     → survived
285     CPK = normal
286     → died
287     CPK = severely-high
288     → died
289     CPK = very-high
290     → survived
291     ejection_fraction = borderline
292     diabetes = no
293     CPK = high
294     platelets = normal
295     age = [40-50]
296     anaemia = no
297     → died
298     anaemia = yes
299     → survived
300     age = [61-70]
301     → survived
302     age = 70plus
303     → survived
304     platelets = low
305     → died
306     CPK = normal
307     → survived
308     CPK = very-high

```

```

309     → survived
310     diabetes = yes
311     anaemia = no
312     age = [61-70]
313     CPK = normal
314     → died
315     CPK = high
316     → survived
317     age = 70plus
318     → survived
319     anaemia = yes
320     → died
321     ejection_fraction = mildly-low
322     high_blood_pressure = yes
323     age = [61-70]
324     anaemia = no
325     → died
326     anaemia = yes
327     → survived
328     age = 70plus
329     CPK = normal
330     → died
331     CPK = very-high
332     → survived
333     CPK = high
334     → survived
335     age = [40-50]
336     → survived
337     age = [51-60]
338     → survived
339     high_blood_pressure = no
340     → survived
341     serum_creatinine = low
342     → survived

```

D MyDT+ diagram (Reduced-Error Pruning)

The pruned tree was obtained on the same training data, with the deterministic 75/25 stratified split between grow and prune sets described in Section 3.2. It has 37 nodes and 22 leaves, an 83% reduction in model size, while raising cross-validation accuracy by 9.5 percentage points.

Listing 4: MyDT+ trained on heart.csv with REP and a 25% prune set.

```

1  serum_creatinine = high
2    sex = M
3      age = [61-70]
4        diabetes = yes
5          → died
6        diabetes = no
7          → survived
8      age = [40-50]
9        serum_sodium = normal
10          CPK = high
11            diabetes = no
12              → died
13            diabetes = yes
14              smoking = yes
15                → died
16              smoking = no
17                → survived
18          CPK = normal
19            anaemia = yes
20              → died
21            anaemia = no
22              → survived
23          CPK = very-high
24            → survived
25        serum_sodium = low
26          → survived
27      age = [51-60]

```

```

28     serum_sodium = low
29     diabetes = yes
30     CPK = normal
31     → died
32     CPK = very-high
33     → survived
34     CPK = high
35     → survived
36     diabetes = no
37     → died
38     serum_sodium = normal
39     CPK = normal
40     anaemia = yes
41     diabetes = no
42     → died
43     diabetes = yes
44     → survived
45     anaemia = no
46     → survived
47     CPK = high
48     → survived
49     CPK = very-high
50     → survived
51     age = 70plus
52     → died
53     sex = F
54     → died
55     serum_creatinine = normal
56     → survived
57     serum_creatinine = low
58     → survived

```